THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — MATHS · NCERT-TYPE CHAPTER TEST SET 3

Chapter 13: **Statistics** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. For the data with classes 50–60, 60–70, 70–80, 80–90, 90–100 with frequencies 5, 10, 5, 3, 2 respectively, The mode of the data is: 1

(a) 65 **(b)** 69.8 **(c)** 70 **(d)** 60

2. For the data with classes 40–50, 50–60, 60–70, 70–80, 80–90, 90–100 with frequencies 4, 7, 8, 8, 12, 11 respectively, The modal class is: 1

(a) 80–90 **(b)** 40–50 **(c)** 60–70 **(d)** 50–60

3. The mean of a distribution with classes 50–60, 60–70, 70–80, 80–90, 90–100, 100–110 and frequencies 12, 18, p, 20, 21, 18 respectively is 82.4. The value of p is: 1

(a) 10 **(b)** 11 **(c)** 13 **(d)** 16

4. The modal class of a frequency distribution is the class: 1

(a) with the lowest frequency **(b)** in the middle of the table
(c) with the highest frequency **(d)** with the highest class mark

5. For the data with classes 10–20, 20–30, 30–40, 40–50, 50–60 with frequencies 7, 2, 4, 8, 4 respectively, The mean of the data is: 1

(a) 36 **(b)** 175 **(c)** 35 **(d)** 34

6. For the data with classes 20–30, 30–40, 40–50, 50–60, 60–70 with frequencies 8, 16, 4, 8, 4 respectively, The median of the data is: 1

(a) 41 **(b)** 40 **(c)** 37.5 **(d)** 30

7. For the data with classes 30–40, 40–50, 50–60, 60–70, 70–80 with frequencies 3, 2, 12, 15, 8 respectively, The mode of the data is: 1

(a) 60.75 **(b)** 70 **(c)** 63 **(d)** 60

8. For the data with classes 50–60, 60–70, 70–80, 80–90, 90–100, 100–110 with frequencies 4, 18, 12, 5, 9, 2 respectively, The modal class is: 1

(a) 60–70 **(b)** 50–60 **(c)** 70–80 **(d)** 80–90

9. The mean of a distribution with classes 10–20, 20–30, 30–40, 40–50, 50–60 and frequencies 7, p, 4, 8, 4 respectively is 35. The value of p is: 1

(a) 1 **(b)** 7 **(c)** 2 **(d)** 4

10. If the assumed mean is a and the deviations $d_i = x_i - a$ are used, then $\bar{x} =$: 1

(a) $a - \dfrac{\sum f_i d_i}{\sum f_i}$ **(b)** $\dfrac{\sum f_i d_i}{\sum f_i}$ **(c)** $a \sum f_i d_i$ **(d)** $a + \dfrac{\sum f_i d_i}{\sum f_i}$

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. The monthly pocket money (in ₹) of students is given below. 2

Pocket money (₹)	70–80	80–90	90–100	100–110	110–120	120–130
Number of students	4	16	4	8	15	3

Find the median.

12. The weights (in kg) of students of a class are given below. 2

Weight (kg)	60–70	70–80	80–90	90–100	100–110
Number of students	5	14	8	2	11

Find the mean.

13. The distances (in km) travelled to office by employees of a company are given below. **2**

Distance (km)	30–40	40–50	50–60	60–70	70–80
Number of employees	7	9	12	14	8

Find the mode.

14. The marks obtained by students in a test are given below. **2**

Marks	70–80	80–90	90–100	100–110	110–120
Number of students	5	8	15	14	8

Find the median.

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The daily electricity consumption (in units) of households in a colony is given below. **3**

Units consumed	70–80	80–90	90–100	100–110	110–120
Number of households	13	16	7	7	7

Find the mean.

16. The distances (in km) travelled to office by employees of a company are given below. **3**

Distance (km)	70–80	80–90	90–100	100–110	110–120
Number of employees	3	7	20	7	13

Find the median.

17. The marks obtained by students in a test are given below. **3**

Marks	10–20	20–30	30–40	40–50	50–60
Number of students	7	2	4	8	4

Find the mode.

OR

The lifetimes (in hours) of electric bulbs tested in a laboratory are given below. **3**

Lifetime (hours)	50–60	60–70	70–80	80–90	90–100	100–110
Number of bulbs	8	10	14	2	13	3

Find the mean.

18. The monthly pocket money (in ₹) of students is given below. **3**

Pocket money (₹)	0–10	10–20	20–30	30–40	40–50
Number of students	4	15	4	4	13

Find the median.

OR

The daily wages (in ₹) of workers in a factory are given below. **3**

Daily wage (₹)	40–50	50–60	60–70	70–80	80–90	90–100
Number of workers	4	3	2	4	8	4

Find the mode.

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. The mean of a distribution with classes 20–30, 30–40, 40–50, 50–60, 60–70 and frequencies 8, 16, 4, p, 4 respectively is 41. Find the value of p. **5**

OR

The median of a distribution with classes 40–50, 50–60, 60–70, 70–80, 80–90, 90–100, 100–110 and frequencies 14, 16, x, 20, y, 16, 8 respectively, and total frequency 100, is 78. Find the value of x. **5**

20. The mean of a distribution with classes 40–50, 50–60, 60–70, 70–80, 80–90 and frequencies 9, 4, 10, 13, p respectively is 64.75. Find the value of p. **5**

OR

The median of a distribution with classes 0–10, 10–20, 20–30, 30–40, 40–50, 50–60 and frequencies 3, 10, x, 5, y, 5 respectively, and total frequency 50, is 32. Find the value of x. **5**

END OF QUESTION PAPER — Total: 40 Marks